A Two-Tier Nonlinear Activity Transfer Criterion

Exact comparison and collar budgets after dependent Gaussian control

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Abstract

We combine three restricted mechanisms developed after the frustrated-boundary obstruction: conditional concentration under coherent staples, exact charging of center frustration, and a dependent Gaussian bad-set theorem. For each connected plaquette support X, split the nonlinear bad event according to a coherent-collar event G_X. If the good channel costs at most a Radon–Nikodym factor $e^{\kappa|X|}$ relative to the Gaussian rate J, and the collar-failure channel costs $e^{-K|X|}$, then the complete connected-event envelope has weight $q_{\text{eff}} = e^{\kappa-J} + e^{-K}$. We prove the resulting rooted event criterion and calculate the exact remaining budget at the frozen ultraviolet witness. The abstract transfer closes for $\kappa = 0.2$ and $K = 7.0$; proving those inequalities for the complete nonlinear SU(2) measure remains open.

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1 Two channels and their information content

Let B_X denote the event that every plaquette in a connected support X is bad at radius ρ. Let G_X be the gauge-covariant coherent-collar event defined from the surrounding staple family. The decomposition

$$B_X = (B_X \cap G_X) \,\dot\cup\, (B_X \cap G_X^c)$$

retains the distinction erased by arbitrary microscopic conditioning.

The dependent Gaussian theorem supplies a per-plaquette rate J. The two nonlinear proof obligations are

$$\mu(B_X \cap G_X) \le e^{\kappa|X|} e^{-J|X|}. \tag{1}$$

$$\mu(B_X \cap G_X^c) \le e^{-K|X|}. \tag{2}$$

The first measures nonlinear action, Haar, and coordinate comparison on a good collar. The second charges the cause of collar failure, including center and topological labels, to the same support.

2 Transfer theorem

Theorem 2.1 (Two-tier connected-event envelope)**.** *If* (1)–(2) *hold for every connected* X*, then*

$$\mu(B_X) \le q_{\mathrm{eff}}^{|X|}, \qquad q_{\mathrm{eff}} = e^{\kappa - J} + e^{-K}. \tag{3}$$

If the plaquette adjacency degree is at most Δ *and*

$$e^{\alpha}\Delta^2 q_{\mathrm{eff}} < 1, \tag{4}$$

then

$$\sup_{p} \sum_{\substack{X \ni p \\ X \text{ connected}}} e^{\alpha|X|}\mu(B_X) \le \frac{e^{\alpha} q_{\mathrm{eff}}}{1 - e^{\alpha}\Delta^2 q_{\mathrm{eff}}}. \tag{5}$$

Proof. The disjoint split and (1)–(2) give

$$\mu(B_X) \le a^{|X|} + b^{|X|} \le (a+b)^{|X|}, \qquad a = e^{\kappa - J}, \quad b = e^{-K},$$

which proves (3). Bound the number of rooted connected sets of size n by $\Delta^{2(n-1)}$ and sum the geometric series. This gives (5) under (4). □

The critical effective weight is

$$q_{\mathrm{crit}} = \frac{e^{-\alpha}}{\Delta^2}. \tag{6}$$

If collar failure were absent, the maximum comparison loss permitted by the counting criterion would be

$$\kappa_{\max} = J + \log q_{\mathrm{crit}}. \tag{7}$$

At fixed κ with $e^{\kappa - J} < q_{\mathrm{crit}}$, the collar rate must obey

$$K > -\log\bigl(q_{\mathrm{crit}} - e^{\kappa - J}\bigr). \tag{8}$$

3 Frozen nonlinear budget

Insert the dependent Gaussian rate $J = 6.69010106986$, $\alpha = 0.1$, and $\Delta = 18$. For the declared targets $\kappa = 0.2$ and $K = 7.0$:

quantity	value
good-channel weight $e^{\kappa - J}$	0.00151839554
collar weight e^{-K}	0.000911881966
q_{eff}	0.00243027750
q_{crit}	0.00279270808
weight slack	0.000362430579
branching ratio	0.870222534
rooted sum upper	0.0206959813
$\kappa_{\max}$ without collar failure	0.809357554
minimum K at $\kappa = 0.2$	6.66534843

Thus the declared pair is strictly sufficient for this counting theorem. It is a target budget, not a measured or proved property of the nonlinear link measure.

4 Finite multiscale composition

At scales $j = 1, \dots, N$, define

$$q_j = e^{\kappa_j - J_j} + e^{-K_j}, \qquad r_j = e^{\alpha_j} \Delta_j^2 q_j.$$

If every $r_j < 1$, then the declared finite-registry exceptional error is bounded by

$$\sum_{j=1}^{N} \frac{e^{\alpha_j} q_j}{1 - r_j}. \tag{9}$$

For the four-scale illustrative registry in the executable calibration, this sum is 0.0286463. Equation (9) is a composition lemma. It does not construct RG maps, prove that the registry follows the Yang–Mills flow, or establish infinite-scale summability.

5 Exact next theorem and frontier audit

The first unproved analytic implication has become the following pair of inequalities, for the complete finite-volume heat-kernel SU(2) measure, uniformly in volume and admitted exterior gauge-invariant data:

$$\mu(B_X \cap G_X) \le e^{0.2|X|} \mu_{\mathrm{G}}(B_X^{\mathrm{lin}}),$$
$$\mu(B_X \cap G_X^c) \le e^{-7|X|}.$$

They must use the same t, ρ, η, block geometry, and support. The collar channel must retain center flux, boundary holonomy, orbit-atlas failure, and topology rather than optimizing them away.

The theorem bounds connected bad events. To obtain a partition-function cluster expansion, one must additionally construct the relevant nonlinear polymers or prove a stochastic domination theorem that legitimately uses this event envelope. No Mayer/Ursell activity identification is asserted here.

The transfer theorem and its budget are C2 results. Nonlinear full-measure control, repeated RG, continuum coherence, Osterwalder–Schrader reconstruction, and the mass gap remain open.